ACADEMIC PRESENTATION / CURATED PUBLIC EXCERPT

Quantum Communication Networks

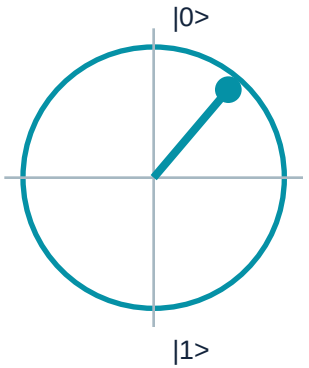
M.Sc. QTEC / Academic presentation / 2025

SCIENTIFIC QUESTION

How do qubit states, entanglement and the CHSH game inform communication protocols?

KEY POINTS

- Bloch vectors represent pure and mixed single-qubit states.
- Tomography uses statistics from measurements in multiple bases.
- CHSH contrasts classical and entanglement-assisted correlations.



MY CONTRIBUTION

Prepared and presented a synthesis of quantum-state concepts, CHSH, dense coding, teleportation and multiple-access models.

LIMITS & ATTRIBUTION

Conceptual and literature-based presentation; no physical tomography, Bell measurement or measured network gain.

Established quantum information theory; course presentation sources.

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